COMPUTATIONAL COUPLING — Experimental Hypotheses

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> **Status: planning document, superseded.** This was written at the start of the project (September 2026), before any experiment was run, and it records the research plan and novelty analysis as they stood then. It is **not** a statement of findings. Several of its premises did not survive: the human H01 microscale tier was withdrawn when its connectivity graph turned out to be empty, the CoupledEventNet / neuromorphic-efficiency direction was withdrawn along with the fabricated silicon figures supporting it, and the central motif hypothesis it sets out was tested and came out with the opposite sign.
>
> For what the project actually found, see `../info.md` and the manuscript. For what was withdrawn and why, see `../AUDIT.md`. This file is kept because the retraction is only checkable against the plan it replaced.

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Hypothesis 1: Functional Superiority of Conserved Motifs

Conserved structural motifs (specifically Type-1 Coherent Feed-Forward Loops and Reciprocal Inhibitory Feedback) identified across C. elegans, Drosophila, and mouse cortical connectomes exhibit statistically significant advantages in temporal coincidence detection, dynamic range, and persistence over degree-matched random configuration null models under event-driven spiking dynamics.

- **Null Hypothesis ($H_{1,0}$):** The dynamical performance (signal-to-noise ratio, temporal memory capacity, spike efficiency) of empirical motifs is statistically indistinguishable from degree-preserving randomized graphs ($p \geq 0.05$, two-tailed permutation test).
- **Falsification Criterion:** If randomized null graphs achieve equal or superior dynamical memory capacity and coincidence selectivity, H_1 is rejected.

Hypothesis 2: Structural Degeneracy & The Remodeling Frontier

Biological connectome subcircuits exhibit a structural degeneracy regime under biological constraints: an empirical circuit G_{orig} can be topologically rewired by a graph edit distance $\Delta_{struct} \geq 0.35$ (35% synaptic turnover) while preserving $\geq 95\%$ of baseline task performance ($F_{task}(G') \geq 0.95 F_{task}(G_{orig})$), provided Dale's E/I sign segregation and core feedback invariants are preserved.

- **Null Hypothesis ($H_{2,0}$):** Any topological remodeling beyond $\Delta_{\text{struct}} > 0.10$ results in catastrophic degradation of task accuracy ($F_{\text{task}} < 0.80 F_{\text{task}}(G_{\text{orig}})$).
- **Falsification Criterion:** If the multi-objective Pareto front collapses immediately upon structural modification under biological constraints, H_2 is rejected.

Hypothesis 3: Bimodal Causal Invariance vs. Developmental Scaffolding

Synaptic connections in empirical connectome subcircuits do not contribute uniformly to dynamic task computation; rather, their causal importance distribution is bimodal: a small subset of connections embedded in conserved motifs act as "computational invariants" (lesioning causes $> 50\%$ performance collapse), whereas the majority of connections represent "developmental scaffolding" that can be pruned or rewired with $< 2\%$ performance impact.

- **Null Hypothesis ($H_{3,0}$):** The distribution of causal lesion impacts across all synapses is unimodal or uniform, showing no segregation between motif-embedded connections and peripheral connections.
- **Falsification Criterion:** If lesioning motif connections versus non-motif connections shows no statistically significant difference in performance collapse, H_3 is rejected.

Hypothesis 4: Neuromorphic Event-Driven Efficiency of CoupledEventNet

An event-driven neural architecture synthesized from cross-species computational invariants (CoupledEventNet / CEN) will achieve at least a $3\times$ reduction in estimated Synaptic Operations (SynOps) and event count compared to conventional dense SNN baselines (standard LIF networks) and connectome reservoirs (`conn2res`) at equivalent or superior temporal classification accuracy on sequential benchmarks.

- **Null Hypothesis ($H_{4,0}$):** CoupledEventNet requires equal or greater SynOps / spike events, or suffers a statistically significant decrease in accuracy ($> 5\%$ drop) relative to standard SNN/LIF baselines.
- **Falsification Criterion:** If CEN does not demonstrate a statistically significant reduction in SynOps and event count at matched accuracy ($p < 0.01$), H_4 is rejected.

Hypothesis 5: Human Micro-Connectome Motif Invariance (H01 Cross-Validation)

The causal computational motifs validated in C. elegans, Drosophila, and mouse cortex (specifically Feed-Forward Coincidence Loops and Reciprocal Inhibitory Feedback) will exhibit statistically significant overrepresentation in the H01 human cortical micro-connectome relative to degree-preserving randomized null models ($Z > 2.5$, $p < 0.01$), confirming their role as cross-scale biological invariants.

- **Null Hypothesis ($H_{5,0}$):** Motif frequencies in H01 human temporal cortex do not differ significantly from randomized configuration nulls, or display a fundamentally divergent motif significance profile unrelated to non-human models.
- **Falsification Criterion:** If empirical H01 subgraphs show no statistical enrichment of FFL or reciprocal inhibitory motifs ($Z \leq 1.0$), H_5 is rejected.

Hypothesis 6: Scale Separation (Synaptic Micro-Motifs vs. Macro-Scale Routing)

Network motifs that govern event-driven computation at the synaptic micro-scale (H01 / MICrONS / Drosophila) do NOT scale self-similarly to the whole-brain macro-scale (HCP tractography); rather, biological nervous systems exhibit scale separation where microcircuits optimize temporal coincidence and gain control (local FFL / reciprocal inhibition), whereas macro-networks optimize global modular routing and communication latency (rich-club hubs / small-world path length).

- **Null Hypothesis ($H_{6,0}$):** The network motif profile and functional dynamics of whole-brain HCP diffusion tractography are identical to nanoscale synaptic micro-connectomes.
- **Falsification Criterion:** If 3-node motif distributions and dynamical memory profiles are isomorphic between H01 and HCP, H_6 is rejected.